Capstone Project

The goal of this capstone project:

Build a simple Retrieval-Augmented Generation (RAG) system for an e-commerce use case using the Amazon Reviews dataset available in the Hugging Face datasets library.

The project will use Python libraries such as datasets, sentence-transformers, and faiss to create an AI-powered product recommendation and customer query assistant.

Project tasks:

- Load the Amazon Reviews dataset using Python
- Clean and preprocess review text data
- Convert reviews into embeddings using Sentence Transformers
- Store embeddings in a FAISS vector database
- Accept user product-related queries
- Retrieve the most relevant reviews/products using semantic search
- Generate helpful responses based on retrieved results

Tools & Libraries:

- Python
- Pandas
- Hugging Face Datasets
- Sentence Transformers
- FAISS
- NumPy

Outcome:
Learners will understand how to build an end-to-end RAG application in Python for the e-commerce industry using real-world data.

Explanation of Dataset Used: Amazon Polarity Dataset

The project uses the Amazon Polarity dataset available in the Hugging Face datasets library.

About the Dataset:
- It contains customer product reviews collected from Amazon.
- The dataset is widely used for text classification and NLP tasks.
- Each record includes:
1. title of the product review
2. review content
3. sentiment label (positive or negative)

Why this Dataset is Useful:
- Real customer feedback on products
- Covers multiple product categories
- Large amount of text data for AI projects
- Useful for recommendation systems
- Helpful for sentiment analysis
- Ideal for RAG applications using semantic search

How it is Used in This Project:
- Product titles and review text are combined
- Converted into embeddings using Sentence Transformers
- Stored inside FAISS vector database
- When a user asks a query, similar reviews/products are retrieved
- Based on retrieved results, product suggestions can be generated

Example Records:
1. "Excellent headphones with great sound quality."
2. "Laptop battery life is poor and performance is slow."
3. "Comfortable shoes and worth the price."

Dataset Load Command:
```
from datasets import load_dataset
dataset = load_dataset("amazon_polarity")
```

This allows direct loading in Python using the Hugging Face library.

All the best